

DAA OUTPUT :

1).Implementation and Performance Analysis of Interpolation Search

```
Array: [2, 5, 10, 15, 23, 35, 48, 60, 75, 90, 105, 120]
Searching for: 35
Found at index: 5, Comparisons: 3
```

Size	IS Time(ms)	BS Time(ms)	IS Comparisons	BS Comparisons
1000	0.0012	0.0014	3	9
5000	0.0012	0.0017	3	12
10000	0.0015	0.0016	4	11
50000	0.0014	0.0022	4	16
100000	0.0049	0.0037	6	12

2).Comparative Analysis of Naive, Rabin-Karp, and KMP Algorithms for String Matching

```
Text: AABAACAADAABAABA
Pattern: AABA
```

```
Naive -> Matches at: [0, 9, 12], Comparisons: 30
```

```
KMP -> Matches at: [0, 9, 12], Comparisons: 18
```

```
RK -> Matches at: [0, 9, 12], Comparisons: 12
```

Pattern	Naive	KMP	RK
AB	12470	10000	1234
ABCD	13234	10000	212
ABCDAB	13279	10011	159
ABCDABCD	13281	10011	136

3).Implementation of Kruskal's and Prim's Algorithms for Minimum Spanning Tree

```
=== Kruskal's MST ===
Edge (0 - 3) Weight: 5
Edge (2 - 4) Weight: 5
Edge (3 - 5) Weight: 6
Edge (0 - 1) Weight: 7
Edge (1 - 4) Weight: 7
Edge (4 - 6) Weight: 9
Total MST Cost: 39

=== Prim's MST ===
Edge (0 - 3) Weight: 5
Edge (3 - 5) Weight: 6
Edge (0 - 1) Weight: 7
Edge (1 - 4) Weight: 7
Edge (4 - 2) Weight: 5
Edge (4 - 6) Weight: 9
Total MST Cost: 39
```

4).Implementation of Single Source Shortest Path Algorithm (Dijkstra's)

Shortest paths from vertex 0:		
Vertex	Distance	Path
0	0	0
1	3	0 -> 2 -> 1
2	1	0 -> 2
3	4	0 -> 2 -> 1 -> 3
4	7	0 -> 2 -> 1 -> 3 -> 4
5	9	0 -> 2 -> 1 -> 3 -> 4 -> 5

5).To find the MIN-Max by applying Divide and Conquer technique

Array: [3, 1, 7, 4, 9, 2, 8, 5, 6, 0]

Min: 0, Max: 9

D&C Comparisons: 14

Naive Comparisons: 18

Size	DC Comps	Naive Comps	Formula $3n/2-2$
10	14	18	13
100	162	198	148
1000	1510	1998	1498
10000	15902	19998	14998